

Strategic Disclosure Papers

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Preface

This book collects five short, self-contained papers on strategic disclosure. The first two chapters are adapted from the supplied regression sources in **Sources/**, while the remaining chapters extend the same disclosure framework to online and multiclass settings. Each chapter stands on its own and focuses only on its own decision or learning problem.

Chapter 1

Regression Decision

Abstract

We study regression with strategic all-or-nothing feature disclosure. The receiver commits to a linear predictor, the sender chooses whether to reveal features, and a channel erases revealed features with probability q . This chapter gives complete derivations for the mandatory and voluntary risk formulas, the sender's best response, and the sufficient and necessary conditions under which voluntary disclosure strictly improves the receiver's risk.

1.1 Problem Setup

Let $X \in \mathbb{R}^d$ be the feature vector, $Y \in \mathbb{R}$ the ground truth, and $U \in \mathbb{R}$ the sender's preferred outcome. We assume

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[Y] = 0, \quad \mathbb{E}[U] = 0.$$

The sender can either reveal the full feature vector X or withhold it, which we denote by $\emptyset$. If the sender reveals X , then the channel erases it with probability $q \in (0, 1)$. The receiver commits to the linear rule

$$r_b(\emptyset) = 0, \quad r_b(x) = b^\top x.$$

The loss is squared error, $\ell(a, b) = (a - b)^2$.

For a fixed receiver parameter b , define the two key statistics

$$F_b(u, x) = (2u - b^\top x)b^\top x, \quad G_b(x, y) = (2y - b^\top x)b^\top x.$$

The sender's best response depends on F_b , and the receiver's gain from a successful revelation is measured by G_b .

Let

$$b_V = \text{Var}(X)^{-1} \text{Cov}(X, Y)$$

be the ordinary least-squares weight, and define the mandatory and strategic optimal risks by

$$R_V^*(q) = \min_b R_V(b, q), \quad R_S^*(q) = \min_b R_S(b, q).$$

The strategic advantage is

$$\Delta(E) = R_V^*(q) - R_S^*(q).$$

1.2 Algebraic Identities

The whole analysis rests on two identities. First,

$$G_b(X, Y) = Y^2 - (Y - b^\top X)^2.$$

Second,

$$G_b(X, Y) - F_b(U, X) = 2(Y - U)b^\top X.$$

The first identity measures how much prediction error is reduced when X is revealed. The second identity measures how far the sender's preference U is from the ground truth Y along the current prediction direction.

1.3 Mandatory Disclosure

Lemma 1.1 (Vanilla risk). *For any $b \in \mathbb{R}^d$, the risk under mandatory disclosure is*

$$R_V(b, q) = \text{Var}(Y) - (1 - q) \text{Var}(b_V^\top X) + (1 - q) \text{Var}((b - b_V)^\top X).$$

In particular,

$$R_V^*(q) = \text{Var}(Y) - (1 - q) \text{Var}(b_V^\top X).$$

Proof. Under mandatory disclosure, the sender always reveals X . The channel erases the feature with probability q , so the receiver incurs loss Y^2 when erasure occurs and loss $(Y - b^\top X)^2$ otherwise. Therefore

$$R_V(b, q) = q\mathbb{E}[Y^2] + (1 - q)\mathbb{E}[(Y - b^\top X)^2].$$

Because Y has mean zero, $\mathbb{E}[Y^2] = \text{Var}(Y)$.

To expand the second term, write

$$Y - b^\top X = (Y - b_V^\top X) + (b_V - b)^\top X.$$

Squaring and taking expectations gives

$$\mathbb{E}[(Y - b^\top X)^2] = \mathbb{E}[(Y - b_V^\top X)^2] + \mathbb{E}[(b_V - b)^\top X]^2 + 2\mathbb{E}[(Y - b_V^\top X)((b_V - b)^\top X)].$$

The cross term is zero by the orthogonality condition for least squares:

$$\mathbb{E}[(Y - b_V^\top X)X] = 0.$$

Hence the cross term vanishes after multiplying by $(b_V - b)^\top$. Also,

$$\mathbb{E}[(Y - b_V^\top X)^2] = \text{Var}(Y) - \text{Var}(b_V^\top X)$$

because both Y and $b_V^\top X$ have mean zero. Finally,

$$\mathbb{E}[(b_V - b)^\top X]^2 = \text{Var}((b - b_V)^\top X).$$

Substituting these identities into the risk formula yields the stated expression. The minimum is attained at $b = b_V$ because the additional variance term is nonnegative and is zero only at b_V . $\square$

1.4 Sender Best Response

Lemma 1.2 (Best response). *For a fixed receiver strategy r_b , the sender reveals X if and only if*

$$F_b(u, x) > 0.$$

Equivalently, the sender withholds when $F_b(u, x) \leq 0$.

Proof. Fix a realization (u, x) . If the sender withholds, the receiver observes $\emptyset$ and predicts 0, so the sender's loss is

$$\ell(u, 0) = u^2.$$

If the sender reveals x , then the channel keeps the feature with probability $1 - q$ and erases it with probability q . The expected sender loss from revealing is therefore

$$q u^2 + (1 - q)(u - b^\top x)^2.$$

The sender prefers revealing exactly when

$$u^2 > q u^2 + (1 - q)(u - b^\top x)^2.$$

Since $1 - q > 0$, this is equivalent to

$$u^2 > (u - b^\top x)^2.$$

Expanding the square gives

$$u^2 - (u - b^\top x)^2 = 2u b^\top x - (b^\top x)^2 = (2u - b^\top x)b^\top x.$$

Hence the sender reveals if and only if $F_b(u, x) > 0$. We break ties in favor of withholding, so $F_b(u, x) = 0$ leads to $\emptyset$. $\square$

1.5 Strategic Risk

Lemma 1.3 (Strategic risk). *For any $b \in \mathbb{R}^d$, the strategic risk is*

$$R_S(b, q) = \text{Var}(Y) - (1 - q)\mathbb{E}[G_b(X, Y) \mathbf{1}\{F_b(U, X) > 0\}].$$

Proof. By the best-response lemma, the sender withholds exactly when $F_b(U, X) \leq 0$. Condition on this event. If $F_b(U, X) \leq 0$, then the receiver observes $\emptyset$ and incurs loss Y^2 . If $F_b(U, X) > 0$, then the sender reveals X ; the channel erases it with probability q , again causing loss Y^2 , and otherwise the receiver incurs $(Y - b^\top X)^2$.

Therefore

$$R_S(b, q) = \mathbb{E}\left[\mathbf{1}\{F_b(U, X) \leq 0\}Y^2 + \mathbf{1}\{F_b(U, X) > 0\}(qY^2 + (1 - q)(Y - b^\top X)^2)\right].$$

Using $\mathbf{1}\{F \leq 0\} + \mathbf{1}\{F > 0\} = 1$, this becomes

$$R_S(b, q) = \mathbb{E}[Y^2] - (1 - q)\mathbb{E}\left[\mathbf{1}\{F_b(U, X) > 0\}(Y^2 - (Y - b^\top X)^2)\right].$$

Because $\mathbb{E}[Y^2] = \text{Var}(Y)$ and

$$Y^2 - (Y - b^\top X)^2 = (2Y - b^\top X)b^\top X = G_b(X, Y),$$

the stated formula follows. $\square$

1.6 Sufficient Condition

Theorem 1.4 (Sufficient condition). *If there exists $b \in \mathbb{R}^d$ such that*

$$\|U - Y\|_2 < \frac{\mathbb{E}|G_b(X, Y)| - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X)}{4\sqrt{\text{Var}(b^\top X)}},$$

then voluntary disclosure is strictly better than mandatory disclosure, i.e.

$$R_S^*(q) < R_V^*(q).$$

Proof. Define

$$\Delta(b, q) = R_S(b, q) - R_V^*(q).$$

Using the previous two lemmas,

$$\Delta(b, q) = (1 - q) \left(\text{Var}(b_V^\top X) - \mathbb{E}[G_b(X, Y) \mathbf{1}\{F_b(U, X) > 0\}] \right).$$

Now write

$$\mathbb{E}[G_b \mathbf{1}\{F_b > 0\}] = \mathbb{E}[G_b] - \mathbb{E}[G_b \mathbf{1}\{F_b \leq 0\}].$$

Also,

$$\mathbb{E}[G_b] = \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X)$$

by the same least-squares identity used in the vanilla-risk proof. Hence

$$\Delta(b, q) = (1 - q) \left(\text{Var}((b - b_V)^\top X) + \mathbb{E}[G_b(X, Y) \mathbf{1}\{F_b(U, X) \leq 0\}] \right).$$

So it is enough to show that the bracketed term is negative.

We split the expectation into a part controlled by the sign of G_b and a part controlled by the mismatch between F_b and G_b :

$$\mathbb{E}[G_b \mathbf{1}\{F_b \leq 0\}] = \mathbb{E}[G_b \mathbf{1}\{G_b \leq 0\}] + \mathbb{E}[G_b(\mathbf{1}\{F_b \leq 0\} - \mathbf{1}\{G_b \leq 0\})].$$

For the first term,

$$\mathbb{E}[G_b \mathbf{1}\{G_b \leq 0\}] = \frac{\mathbb{E}[G_b] - \mathbb{E}|G_b|}{2}.$$

Substituting the identity for $\mathbb{E}[G_b]$ gives

$$\mathbb{E}[G_b \mathbf{1}\{G_b \leq 0\}] = -\frac{1}{2}\mathbb{E}|G_b| + \frac{1}{2}\text{Var}(b_V^\top X) - \frac{1}{2}\text{Var}((b - b_V)^\top X).$$

For the second term, the integrand is nonzero only when F_b and G_b have opposite signs. On that event,

$$|G_b| \leq |G_b - F_b|.$$

But

$$G_b(X, Y) - F_b(U, X) = 2(Y - U)b^\top X,$$

so

$$|G_b - F_b| = 2|Y - U| |b^\top X|.$$

Therefore,

$$\mathbb{E}[G_b(\mathbf{1}\{F_b \leq 0\} - \mathbf{1}\{G_b \leq 0\})] \leq 2\mathbb{E}[|Y - U| |b^\top X|].$$

Applying Cauchy-Schwarz gives

$$2\mathbb{E}[|Y - U| |b^\top X|] \leq 2\|U - Y\|_2 \sqrt{\text{Var}(b^\top X)}.$$

Combining the two bounds yields

$$\mathbb{E}[G_b \mathbf{1}\{F_b \leq 0\}] \leq -\frac{1}{2}\mathbb{E}|G_b| + \frac{1}{2}\text{Var}(b_V^\top X) - \frac{1}{2}\text{Var}((b - b_V)^\top X) + 2\|U - Y\|_2 \sqrt{\text{Var}(b^\top X)}.$$

Substitute this into the expression for $\Delta(b, q)$:

$$\Delta(b, q) \leq (1 - q) \left[\frac{1}{2}\text{Var}(b_V^\top X) + \frac{1}{2}\text{Var}((b - b_V)^\top X) - \frac{1}{2}\mathbb{E}|G_b| + 2\|U - Y\|_2 \sqrt{\text{Var}(b^\top X)} \right].$$

Because $1 - q > 0$, the sign of $\Delta(b, q)$ is controlled by the bracketed term. If

$$4\|U - Y\|_2 \sqrt{\text{Var}(b^\top X)} < \mathbb{E}|G_b| - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X),$$

then the bracket is negative, so $\Delta(b, q) < 0$. This means $R_S(b, q) < R_V^*(q)$ for that b , and therefore $R_S^*(q) < R_V^*(q)$. $\square$

1.7 Necessary Condition

Theorem 1.5 (Necessary condition). *If voluntary disclosure is strictly better for some receiver strategy, then there exists $b \in \mathbb{R}^d$ such that*

$$\mathbb{E}|G_b(X, Y)| - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) > 0.$$

Proof. We again start from

$$\Delta(b, q) = (1 - q) \left(\text{Var}((b - b_V)^\top X) + \mathbb{E}[G_b(X, Y) \mathbf{1}\{F_b(U, X) \leq 0\}] \right).$$

To lower bound the expectation term, compare it with the event defined by the sign of G_b . Pointwise,

$$G_b \mathbf{1}\{F_b \leq 0\} \geq G_b \mathbf{1}\{G_b \leq 0\}.$$

This is immediate by cases. If $G_b \leq 0$, then the left-hand side is either G_b or 0, both of which are at least $G_b = G_b \mathbf{1}\{G_b \leq 0\}$. If $G_b > 0$, then the right-hand side is 0 and the left-hand side is nonnegative, so the inequality still holds. Taking expectations gives

$$\mathbb{E}[G_b \mathbf{1}\{F_b \leq 0\}] \geq \mathbb{E}[G_b \mathbf{1}\{G_b \leq 0\}].$$

Now

$$\mathbb{E}[G_b \mathbf{1}\{G_b \leq 0\}] = \frac{\mathbb{E}[G_b] - \mathbb{E}|G_b|}{2} = \frac{\text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) - \mathbb{E}|G_b|}{2}.$$

Therefore

$$\Delta(b, q) \geq (1 - q) \left[\text{Var}((b - b_V)^\top X) + \frac{\text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) - \mathbb{E}|G_b|}{2} \right],$$

which simplifies to

$$\Delta(b, q) \geq \frac{1 - q}{2} \left(\text{Var}(b_V^\top X) + \text{Var}((b - b_V)^\top X) - \mathbb{E}|G_b| \right).$$

Now suppose that for every b the quantity

$$\mathbb{E}|G_b(X, Y)| - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X)$$

is nonpositive. Then the lower bound above is nonnegative for every b , so $\Delta(b, q) \geq 0$ for every b . That means the receiver cannot do better under voluntary disclosure than under mandatory disclosure, so no strategic advantage exists.

Taking the contrapositive, if a strategic advantage does exist, then some b must satisfy

$$\mathbb{E}|G_b(X, Y)| - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) > 0.$$

$\square$

1.8 Remarks

The noise parameter q never changes the sign of the strategic advantage in these bounds. It only multiplies the final difference by the positive factor $1 - q$.

The necessary condition says that the numerator in the sufficient bound must be positive. If it is not, then the sufficient condition is vacuous and no positive strategic advantage can be certified.

1.9 Conclusion

The regression decision problem can be reduced to a small set of algebraic claims. The mandatory regime gives the classical least-squares decomposition, the sender's best response is a sign test on F_b , and the comparison between F_b and G_b yields both the sufficient and necessary conditions for voluntary disclosure to help.

Chapter 2

Regression Offline Learning

Abstract

We study the batch-learning version of strategic regression. The receiver observes a finite i.i.d. sample from an unknown environment and must choose both a disclosure regime and a linear predictor. This chapter derives the empirical objectives, proves uniform generalization bounds for the vanilla and strategic risks, and then shows that the empirical regime-selection rule achieves near-oracle excess risk.

2.1 Setup

Let $S = \{(u_i, x_i, y_i)\}_{i=1}^n$ be i.i.d. samples drawn from an unknown distribution on (U, X, Y) . We assume the boundedness conditions

$$\|x_i\|_\infty \leq C, \quad |u_i| \leq C, \quad |y_i| \leq C,$$

and restrict the receiver to linear predictors

$$r_b(\emptyset) = 0, \quad r_b(x) = b^\top x, \quad \|b\|_\infty \leq C.$$

As in the decision problem, define

$$F_b(u, x) = (2u - b^\top x)b^\top x, \quad G_b(x, y) = (2y - b^\top x)b^\top x.$$

The receiver compares two disclosure regimes:

- mandatory disclosure, where the sender always reveals features;
- voluntary disclosure, where the sender follows the strategic best response induced by b .

For the batch problem we use the empirical risks

$$\hat{R}_V(b, q) = \frac{1}{n} \sum_{i=1}^n y_i^2 - (1 - q) \frac{1}{n} \sum_{i=1}^n (y_i - b^\top x_i)^2,$$

and

$$\hat{R}_S(b, q) = \frac{1}{n} \sum_{i=1}^n y_i^2 - (1 - q) \frac{1}{n} \sum_{i=1}^n G_b(x_i, y_i) \mathbf{1}\{F_b(u_i, x_i) > 0\}.$$

Let

$$\hat{b}_V \in \arg \min_b \hat{R}_V(b, q), \quad \hat{b}_S \in \arg \min_b \hat{R}_S(b, q).$$

The regime-selection algorithm outputs the regime with the smaller empirical risk.

2.2 Empirical Regime Selection

The empirical strategic advantage is

$$\hat{\Delta}(S) = \min_b \hat{R}_V(b, q) - \min_b \hat{R}_S(b, q).$$

If $\hat{\Delta}(S) > 0$, the learner chooses voluntary disclosure; otherwise it chooses mandatory disclosure. The learning goal is to show that this data-driven choice behaves like the oracle choice that knows the optimal regime in advance.

To prove this, we need three ingredients:

1. a uniform bound for the vanilla risk;
2. a uniform bound for the strategic risk;
3. a stability argument for the regime-selection step.

2.3 Auxiliary Facts

We use three standard learning-theory facts.

Lemma 2.1 (Linear class complexity). *Let $H = \{h_b : (x, y) \mapsto y - b^\top x : \|b\|_\infty \leq C\}$. Then for any sample Z , the empirical Rademacher complexity satisfies*

$$R(H; Z) = O\left(\frac{dC^2}{\sqrt{n}}\right).$$

Proof. The class is linear in the lifted variables (y, x) . Since $|y| \leq C$ and $|b^\top x| \leq dC^2$, each function in H is uniformly bounded by $C + dC^2$. Standard bounds for bounded linear classes give the stated rate. $\square$

Lemma 2.2 (Quadratic threshold VC dimension). *Let*

$$F = \{f_b : (u, x) \mapsto \mathbf{1}\{(2u - b^\top x)b^\top x > 0\} : \|b\|_\infty \leq C\}.$$

Then

$$d_{VC}(F) \leq \frac{(d+1)(d+2)}{2} = O(d^2).$$

Proof. The decision boundary is quadratic in (u, x) . After lifting to the feature vector containing all linear terms and all quadratic monomials, the class becomes a family of linear halfspaces in a space of dimension $(d+1)(d+2)/2$. The VC dimension of halfspaces is at most the ambient dimension, which yields the claim. $\square$

Lemma 2.3 (Quadratic loss Lipschitz bound). *On the bounded range induced by $|y| \leq C$ and $|b^\top x| \leq dC^2$, the loss $\ell(y, b^\top x) = (y - b^\top x)^2$ is $O(dC^2)$ -Lipschitz in its linear argument.*

Proof. For fixed y , the derivative with respect to $z = b^\top x$ is $-2(y - z)$. On the bounded domain, $|y - z| \leq C + dC^2$, so the derivative is bounded by $2(C + dC^2) = O(dC^2)$. $\square$

2.4 Vanilla Risk Bound

Lemma 2.4 (Uniform vanilla generalization). *With probability at least $1 - \delta$,*

$$\sup_b |R_V(b, q) - \hat{R}_V(b, q)| \leq O\left(\frac{d^2 C^4}{\sqrt{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

Proof. Write

$$R_V(b, q) - \hat{R}_V(b, q) = \left(\text{Var}(Y) - \frac{1}{n} \sum_{i=1}^n y_i^2\right) + (1 - q) \left(\hat{L}_V(b) - L_V(b)\right),$$

where

$$L_V(b) = \mathbb{E}[(Y - b^\top X)^2], \quad \hat{L}_V(b) = \frac{1}{n} \sum_{i=1}^n (y_i - b^\top x_i)^2.$$

The first term is a centered average of bounded random variables in $[0, C^2]$, so Hoeffding's inequality gives

$$\left| \text{Var}(Y) - \frac{1}{n} \sum_{i=1}^n y_i^2 \right| \leq C^2 \sqrt{\frac{\log(2/\delta)}{2n}}$$

with probability at least $1 - \delta/2$.

For the second term, apply the standard Rademacher generalization bound to the loss class

$$\ell \circ H = \{(x, y) \mapsto (y - b^\top x)^2 : \|b\|_\infty \leq C\}.$$

By the Lipschitz lemma and the linear-class complexity bound,

$$R(\ell \circ H; Z) \leq O(dC^2) R(H; Z) = O\left(\frac{d^2 C^4}{\sqrt{n}}\right).$$

Combining this with the concentration term yields the stated bound. $\square$

2.5 Strategic Risk Bound

Lemma 2.5 (Uniform strategic generalization). *With probability at least $1 - \delta$,*

$$\sup_b |R_S(b, q) - \hat{R}_S(b, q)| \leq O\left(d^3 C^4 \sqrt{\frac{\log n}{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

Proof. We decompose the difference as

$$R_S(b, q) - \hat{R}_S(b, q) = \left(\text{Var}(Y) - \frac{1}{n} \sum_{i=1}^n y_i^2\right) + (1 - q) \left(\mathbb{E}[h_b(Z)] - \hat{\mathbb{E}}[h_b(Z)]\right),$$

where $Z = (U, X, Y)$ and

$$h_b(u, x, y) = G_b(x, y) \mathbf{1}\{F_b(u, x) > 0\}.$$

The variance term is the same as in the vanilla proof and therefore contributes

$$O\left(C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

To bound the second term uniformly over b , define the function classes

$$F = \{f_b(u, x) = \mathbf{1}\{F_b(u, x) > 0\} : \|b\|_\infty \leq C\},$$

and

$$G = \{g_b(x, y) = G_b(x, y) : \|b\|_\infty \leq C\}.$$

Then $h_b = f_b \cdot g_b$, so h_b lies in the product class $F \cdot G$.

By the standard Rademacher bound for product classes,

$$R(F \cdot G; Z) \leq C_G R(F; Z) + C_F R(G; Z),$$

where C_F bounds F and C_G bounds G . Since F is indicator-valued, $C_F = 1$. Since

$$|G_b(x, y)| \leq (2|y| + |b^\top x|)|b^\top x| \leq O(d^2 C^4),$$

we may take $C_G = O(d^2 C^4)$.

Next, by the VC-dimension lemma and the standard growth-function bound,

$$R(F; Z) = O\left(\sqrt{\frac{d_{VC}(F) \log n}{n}}\right) = O\left(d\sqrt{\frac{\log n}{n}}\right).$$

Also, by the quadratic-form contraction argument,

$$R(G; Z) = O\left(\frac{d^2 C^4}{\sqrt{n}}\right).$$

Therefore

$$R(F \cdot G; Z) \leq O(d^2 C^4) \cdot O\left(d\sqrt{\frac{\log n}{n}}\right) + O\left(\frac{d^2 C^4}{\sqrt{n}}\right),$$

which is dominated by

$$O\left(d^3 C^4 \sqrt{\frac{\log n}{n}}\right).$$

Applying the standard generalization bound to the product class gives the stated result. $\square$

2.6 Strategic Advantage Estimation

Let

$$\Delta^* = R_V^*(q) - R_S^*(q), \quad \hat{\Delta}(S) = \min_b \hat{R}_V(b, q) - \min_b \hat{R}_S(b, q).$$

Theorem 2.6 (Advantage estimation). *With probability at least $1 - \delta$,*

$$|\Delta^* - \hat{\Delta}(S)| \leq O\left(d^3 C^4 \sqrt{\frac{\log n}{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

Proof. Use the inequality

$$\left| \min_b a_b - \min_b \hat{a}_b \right| \leq \sup_b |a_b - \hat{a}_b|.$$

Applying this to the vanilla and strategic risks gives

$$|\Delta^* - \hat{\Delta}(S)| \leq |R_V^*(q) - \hat{R}_V^*(q)| + |R_S^*(q) - \hat{R}_S^*(q)|.$$

The first term is controlled by the vanilla lemma, and the second is controlled by the strategic lemma. Taking a union bound over the two events yields the stated estimate. $\square$

2.7 Excess Risk of the Selection Algorithm

Define

$$\epsilon(n, \delta) = O\left(d^3 C^4 \sqrt{\frac{\log n}{n}} + C^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

Theorem 2.7 (Offline learning guarantee). *For the regime-selection algorithm defined above, with probability at least $1 - \delta$, the excess risk relative to the oracle disclosure regime is at most $O(\epsilon(n, \delta))$.*

Proof. There are two cases.

First, suppose $|\hat{\Delta}(S)| > \epsilon(n, \delta)$. Then the estimation error cannot flip the sign of the strategic advantage, because

$$|\Delta^* - \hat{\Delta}(S)| \leq \epsilon(n, \delta).$$

Hence the algorithm selects the same disclosure regime as the oracle. The only remaining error is the within-regime generalization gap, which is already bounded by $\epsilon(n, \delta)$.

Second, suppose $|\hat{\Delta}(S)| \leq \epsilon(n, \delta)$. Then

$$|\Delta^*| \leq |\Delta^* - \hat{\Delta}(S)| + |\hat{\Delta}(S)| \leq 2\epsilon(n, \delta).$$

So even if the algorithm picks the wrong regime, the additional penalty for that mistake is at most $2\epsilon(n, \delta)$. Adding the within-regime estimation error gives total excess risk at most $3\epsilon(n, \delta)$.

In either case, the excess risk is $O(\epsilon(n, \delta))$. $\square$

2.8 Conclusion

The offline strategic-regression learner is statistically consistent. The proof breaks into three pieces: a vanilla-risk concentration bound, a strategic-risk concentration bound, and a regime-selection stability argument. Together they show that finite-sample learning recovers the same disclosure choice that an oracle would make up to the stated rate.

Chapter 3

Regression Online Learning

Abstract

We study repeated strategic regression in an online setting. At each round, a receiver publishes a linear prediction rule and a sender decides whether to reveal features based on a private preference outcome. In the full-feedback benchmark, the receiver observes the full triple (u_t, x_t, y_t) and can run online gradient descent directly on the strategic population risk. In the partial-feedback setting, the receiver sees only revealed features and uses a surrogate update. We derive the population gradient, prove the $O(1/\sqrt{T})$ full-feedback rate, and then prove the partial-feedback bound by decomposing the error into surrogate-regret and bias terms.

3.1 Online Model

We work in the noiseless online regime, so the channel itself does not erase revealed features. The only stochasticity comes from the sender's strategic choice.

At round t , the receiver commits to a weight vector $b_t \in B \subset \mathbb{R}^d$ and predicts

$$\hat{y}_{b_t}(x) = b_t^\top x, \quad \hat{y}_{b_t}(\emptyset) = 0.$$

Nature draws a triple

$$z_t = (u_t, x_t, y_t)$$

from an unknown distribution P . The sender compares the gain from revealing x_t against withholding it using

$$F_{b_t}(z_t) = (2u_t - b_t^\top x_t)b_t^\top x_t.$$

The revelation indicator $A_t \in \{0, 1\}$ is sampled as

$$A_t \sim \text{Bernoulli}(\psi(F_{b_t}(z_t))),$$

where $\psi : \mathbb{R} \rightarrow [0, 1]$ is smooth, increasing, and Lipschitz. We write ψ' for its derivative.

The instantaneous quadratic loss is

$$\ell_t(b_t; z_t, A_t) = (1 - A_t)y_t^2 + A_t(y_t - b_t^\top x_t)^2.$$

3.2 Population Risk

The strategic population risk is

$$L(b) = \mathbb{E}_{z_t \sim P, A_t \sim \psi}[\ell_t(b; z_t, A_t)].$$

Taking expectation over A_t gives the more useful form

$$L(b) = \mathbb{E}[y_t^2 - \psi(F_b(z_t)) G_b(z_t)],$$

where

$$G_b(z_t) = (2y_t - b^\top x_t) b^\top x_t.$$

The risk is generally nonconvex because the revelation probability $\psi(F_b(z_t))$ depends on b in a nonlinear way. We therefore assume a local regularity condition around a relevant minimizer.

Assumption 3.1 (Local regularity). *There exists a local minimizer b^* such that, in a neighborhood of b^* ,*

$$\nabla^2 L(b^*) \succeq \lambda I$$

for some $\lambda > 0$, and the stochastic gradients satisfy $\|g_t\| \leq G$ almost surely. Also assume $\|b_1 - b^*\| \leq D$.

3.3 Gradient Derivation

Lemma 3.2 (Population gradient). *The gradient of the population risk is*

$$\nabla L(b) = -2\mathbb{E}[\psi'(F_b(z_t))(u_t - b^\top x_t)G_b(z_t)x_t + \psi(F_b(z_t))(y_t - b^\top x_t)x_t].$$

Proof. Start from

$$L(b) = \mathbb{E}[y_t^2 - \psi(F_b(z_t))G_b(z_t)].$$

The term y_t^2 does not depend on b , so we only differentiate the product $\psi(F_b)G_b$.

Let

$$s = b^\top x_t.$$

Then

$$F_b(z_t) = (2u_t - s)s = 2u_t s - s^2, \quad G_b(z_t) = (2y_t - s)s = 2y_t s - s^2.$$

Since $\nabla_b s = x_t$, we get

$$\nabla_b F_b(z_t) = 2(u_t - s)x_t, \quad \nabla_b G_b(z_t) = 2(y_t - s)x_t.$$

Now apply the product and chain rules:

$$\nabla L(b) = -\mathbb{E}[\psi'(F_b(z_t))\nabla_b F_b(z_t)G_b(z_t) + \psi(F_b(z_t))\nabla_b G_b(z_t)].$$

Substituting the two gradient formulas yields

$$\nabla L(b) = -2\mathbb{E}[\psi'(F_b(z_t))(u_t - b^\top x_t)G_b(z_t)x_t + \psi(F_b(z_t))(y_t - b^\top x_t)x_t],$$

as claimed. $\square$

3.4 Full Feedback

In the full-feedback benchmark, the receiver observes (u_t, x_t, y_t) regardless of whether the sender reveals x_t . This allows an unbiased stochastic gradient estimate of $\nabla L(b_t)$.

Let

$$g_t = -2[\psi'(F_{b_t}(z_t))(u_t - b_t^\top x_t)G_{b_t}(z_t)x_t + \psi(F_{b_t}(z_t))(y_t - b_t^\top x_t)x_t].$$

Then $\mathbb{E}[g_t | b_t] = \nabla L(b_t)$.

The receiver updates with projected online gradient descent:

$$b_{t+1} = \Pi_B(b_t - \eta_t g_t).$$

Theorem 3.3 (Full-feedback convergence). *Under the local regularity assumption above, choosing the constant step size*

$$\eta = \frac{D}{G\sqrt{T}}$$

gives

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[L(b_t) - L(b^*)] = O\left(\frac{1}{\sqrt{T}}\right).$$

Proof. Using the nonexpansiveness of projection,

$$\|b_{t+1} - b^*\|^2 \leq \|b_t - \eta g_t - b^*\|^2.$$

Expanding the right-hand side gives

$$\|b_{t+1} - b^*\|^2 = \|b_t - b^*\|^2 - 2\eta g_t^\top (b_t - b^*) + \eta^2 \|g_t\|^2.$$

Rearranging and taking conditional expectation over the randomness at round t yields

$$2\eta \mathbb{E}[\nabla L(b_t)^\top (b_t - b^*)] \leq \mathbb{E}\|b_t - b^*\|^2 - \mathbb{E}\|b_{t+1} - b^*\|^2 + \eta^2 G^2.$$

Because b_t remains inside the locally convex basin, first-order convexity implies

$$L(b_t) - L(b^*) \leq \nabla L(b_t)^\top (b_t - b^*).$$

Summing from $t = 1$ to T and telescoping the distance terms gives

$$\begin{aligned} 2\eta \sum_{t=1}^T \mathbb{E}[L(b_t) - L(b^*)] &\leq \mathbb{E}\|b_1 - b^*\|^2 - \mathbb{E}\|b_{T+1} - b^*\|^2 + \eta^2 G^2 T \\ &\leq D^2 - \mathbb{E}\|b_{T+1} - b^*\|^2 + \eta^2 G^2 T \\ &\leq D^2 + \eta^2 G^2 T. \end{aligned}$$

Divide by $2\eta T$:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[L(b_t) - L(b^*)] \leq \frac{D^2}{2\eta T} + \frac{\eta G^2}{2}.$$

With $\eta = D/(G\sqrt{T})$, both terms are $O(1/\sqrt{T})$, so the average excess strategic risk decays at rate $O(1/\sqrt{T})$. $\square$

3.5 Partial Feedback

In the partial-feedback setting, the sender's private preference u_t is hidden and the receiver does not observe x_t when $A_t = 0$. The receiver therefore uses a surrogate objective built from the observable label y_t .

Define the surrogate risk by replacing F_b with G_b inside the revelation probability:

$$\tilde{L}(b) = \mathbb{E}[y_t^2 - \psi(G_b(z_t)) G_b(z_t)].$$

The surrogate gradient estimator is

$$\tilde{g}_t = \begin{cases} -2 \left[\psi'(G_{b_t}(z_t))(y_t - b_t^\top x_t) G_{b_t}(z_t) x_t + \psi(G_{b_t}(z_t))(y_t - b_t^\top x_t) x_t \right], & A_t = 1, \\ 0, & A_t = 0. \end{cases}$$

We assume the expected revelation probability is bounded away from zero uniformly over the domain B :

$$p_{\min} = \inf_{b \in B} \mathbb{E}_{z \sim P}[\psi(F_b(z))] > 0.$$

Let $\tilde{b}^*$ be a local minimizer of $\tilde{L}$.

Theorem 3.4 (Partial-feedback convergence). *Assume $\tilde{L}$ is locally λ -strongly convex and L -smooth near $\tilde{b}^*$, and that $p_{\min} > 0$. Then*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[L(b_t) - L(b^*)] = O\left(\frac{1}{\sqrt{T}}\right) + \max\{O(\varepsilon), O(\varepsilon^2)\},$$

where

$$\varepsilon = \|U - Y\|_{L^2}.$$

Proof. We decompose the excess risk at each round as

$$L(b_t) - L(b^*) \leq |L(b_t) - \tilde{L}(b_t)| + [\tilde{L}(b_t) - \tilde{L}(\tilde{b}^*)] + |\tilde{L}(\tilde{b}^*) - L(b^*)|.$$

We bound the three terms separately and then average over t .

1. Landscape bias. The only difference between L and $\tilde{L}$ is that the true revelation statistic uses u_t while the surrogate uses y_t . Since ψ is Lipschitz and all quantities are bounded, the risk map is Lipschitz in this distributional parameter. Therefore

$$|L(b) - \tilde{L}(b)| \leq K\|U - Y\|_{L^2} = K\varepsilon$$

for some constant K depending only on the boundedness and Lipschitz parameters.

2. Surrogate regret. The surrogate update is just projected OGD on $\tilde{L}$, except gradients are available only when $A_t = 1$. Because the expected revelation probability is at least $p_{\min}$, the effective sample size is reduced by at most a factor of $p_{\min}$. Standard OGD analysis therefore gives

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\tilde{L}(b_t) - \tilde{L}(\tilde{b}^*)] = O\left(\frac{1}{p_{\min}\sqrt{T}}\right).$$

3. Minima gap. We compare the optima of L and $\tilde{L}$ by first bounding the distance between the minimizers. Since b^* is a local minimizer of L and $\tilde{b}^*$ is a local minimizer of $\tilde{L}$, we have

$$\nabla L(b^*) = 0, \quad \nabla \tilde{L}(\tilde{b}^*) = 0.$$

Strong convexity of L implies

$$\lambda\|b^* - \tilde{b}^*\| \leq \|\nabla L(\tilde{b}^*) - \nabla L(b^*)\| = \|\nabla L(\tilde{b}^*)\|.$$

But since the gradients ∇L and $\nabla \tilde{L}$ only differ by the replacement of the unobserved preference u_t with the surrogate y_t , and because ψ and its derivative are bounded and Lipschitz with bounded features, the gradient is Lipschitz continuous in this distributional parameter. Thus, the difference is bounded by the distance between U and Y :

$$\|\nabla L(\tilde{b}^*)\| = \|\nabla L(\tilde{b}^*) - \nabla \tilde{L}(\tilde{b}^*)\| \leq K\varepsilon,$$

so

$$\|b^* - \tilde{b}^*\| \leq \frac{K\varepsilon}{\lambda}.$$

By L -smoothness of $\tilde{L}$,

$$|\tilde{L}(\tilde{b}^*) - \tilde{L}(b^*)| \leq \frac{L}{2}\|b^* - \tilde{b}^*\|^2 \leq \frac{LK^2}{2\lambda^2}\varepsilon^2.$$

Adding the two bias terms gives

$$|\tilde{L}(\tilde{b}^*) - L(b^*)| = O(\varepsilon) + O(\varepsilon^2).$$

Combining the three bounds and averaging over t yields

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[L(b_t) - L(b^*)] = O\left(\frac{1}{\sqrt{T}}\right) + \max\{O(\varepsilon), O(\varepsilon^2)\}.$$

This is the stated result. $\square$

3.6 Bandit Optimization for Unknown Preferences

To avoid the irreducible bias ε from the surrogate approach, we can directly optimize the true population risk $L(b)$ using a bandit gradient estimator. We observe that although the receiver does not know the private preference u_t (and thus cannot compute the exact gradient $\nabla \ell_t$), they *can* always compute the realized instantaneous loss:

$$\ell_t(b_t; z_t, A_t) = (1 - A_t)y_t^2 + A_t(y_t - b_t^\top x_t)^2.$$

If $A_t = 0$, the loss is simply y_t^2 , which only requires the publicly known label y_t . If $A_t = 1$, the sender reveals x_t , allowing the receiver to compute $(y_t - b_t^\top x_t)^2$. In both cases, the scalar loss is perfectly observable.

We can therefore apply an online bandit gradient descent algorithm with one-point feedback. At each round t , the receiver draws a random unit vector v_t uniformly from the surface of the unit sphere $\mathbb{S}^{d-1}$. The receiver deploys the perturbed prediction rule

$$\hat{b}_t = b_t + \delta_t v_t,$$

where $\delta_t > 0$ is a small exploration radius. The receiver observes the loss $c_t = \ell_t(\hat{b}_t; z_t, A_t)$ and constructs the bandit gradient estimator:

$$\hat{g}_t = \frac{d}{\delta_t} c_t v_t,$$

where d is the dimension. The parameter is updated as $b_{t+1} = \Pi_B(b_t - \eta_t \hat{g}_t)$.

Theorem 3.5 (Bandit-feedback convergence). *Assume L is locally λ -strongly convex and H -smooth. By running the bandit gradient descent algorithm with learning rate $\eta_t = \frac{1}{\lambda t}$ and exploration radius $\delta_t = \delta_0 t^{-1/4}$, the average regret is bounded by:*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[L(\hat{b}_t) - L(b^*)] = O\left(\frac{d}{\sqrt{T}}\right).$$

Proof. By Stokes' theorem, the expectation of the one-point estimator over the random direction v_t is

$$\mathbb{E}_{v_t} \left[\frac{d}{\delta_t} \ell_t(b_t + \delta_t v_t; z_t, A_t) v_t \right] = \nabla \hat{L}_{\delta_t}(b_t),$$

where $\hat{L}_{\delta_t}(b_t) = \mathbb{E}_{v \sim \mathbb{B}^d} [L(b_t + \delta_t v)]$ is the smoothed risk over the solid unit ball $\mathbb{B}^d$. Thus, $\hat{g}_t$ is an unbiased estimator of the gradient of the smoothed risk.

Because L is H -smooth, the smoothed risk uniformly approximates the true risk to second order:

$$\sup_{b \in B} |\hat{L}_{\delta_t}(b) - L(b)| \leq \frac{H}{2} \delta_t^2.$$

The expected squared norm of our gradient estimator is bounded by:

$$\mathbb{E}[\|\hat{g}_t\|^2] = \mathbb{E}\left[\left\|\frac{d}{\delta_t}c_tv_t\right\|^2\right] \leq \frac{d^2c_{\max}^2}{\delta_t^2},$$

where $c_{\max}$ is the maximum instantaneous loss.

We apply the standard strongly convex online gradient descent regret bound to the smoothed function $\hat{L}_{\delta_t}$, which inherits the λ -strong convexity of L . Using the learning rate $\eta_t = \frac{1}{\lambda t}$, the cumulative expected regret against b^* is:

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}[\hat{L}_{\delta_t}(\hat{b}_t) - \hat{L}_{\delta_t}(b^*)] &\leq \sum_{t=1}^T \frac{\eta_t}{2} \mathbb{E}[\|\hat{g}_t\|^2] \\ &\leq \frac{d^2c_{\max}^2}{2\lambda} \sum_{t=1}^T \frac{1}{\delta_t^2 t}. \end{aligned}$$

To translate this back to the true risk L , we add the approximation error $H\delta_t^2$ for each step. Substituting $\delta_t = \delta_0 t^{-1/4}$, we get:

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}[L(\hat{b}_t) - L(b^*)] &\leq \sum_{t=1}^T \left(\frac{d^2c_{\max}^2}{2\lambda\delta_0^2} t^{-1/2} + H\delta_0^2 t^{-1/2} \right) \\ &= O\left(\left(\frac{d^2}{\delta_0^2} + \delta_0^2\right) \sqrt{T}\right). \end{aligned}$$

By choosing the exploration constant $\delta_0 \propto \sqrt{d}$, the leading term becomes $O(d\sqrt{T})$. Dividing by T yields the average regret bound $O(d/\sqrt{T})$.

We note that for general non-linear convex functions, one-point bandit feedback typically suffers a slower $O(T^{-1/4})$ average regret. However, the combination of local strong convexity and smoothness allows the faster $O(1/\sqrt{T})$ rate to be recovered.

Comparing this to the partial feedback surrogate approach, the convergence rate trades the constant factor 1 for the dimension d , but critically avoids the asymptotic plateau. That is, the surrogate bound converges to a nonzero bias floor $\max\{O(\varepsilon), O(\varepsilon^2)\}$, whereas the bandit method strictly converges to the true optimum with an average error of $O(d/\sqrt{T})$, fully eliminating the need to know the alignment between U and Y . $\square$

3.7 Conclusion

The online regression problem remains learnable under strategic disclosure. Full feedback reduces to standard OGD on the strategic population risk, while partial feedback with a surrogate incurs an additional bias from the hidden preference signal. By utilizing a one-point bandit gradient estimator, this irreducible bias can be completely eliminated at the cost of a dimension-dependent variance factor. Ultimately, the problem offers a clear trade-off between the convergence rate scaling and the asymptotic bias.

Chapter 4

Multiclass Classification Decision

Abstract

We study strategic feature withholding in multiclass classification under 0-1 loss. The receiver commits to a default class for empty messages and uses a Bayes classifier when features are revealed. The sender has a private preferred class and chooses whether to reveal features based on the receiver's committed default. This chapter derives the sender's best response, the exact strategic risk for each default class, and the piecewise-linear structure of the optimal risk as a function of the channel noise.

4.1 Problem Setup

Let (U, X, Y) be a random triple, where $X \in \mathbb{R}^d$ is the feature vector, $Y \in \{1, \dots, K\}$ is the ground-truth class, and $U \in \{1, \dots, K\}$ is the sender's preferred class.

We evaluate classification using the 0-1 loss

$$\ell(y, \hat{y}) = \mathbf{1}\{\hat{y} \neq y\}.$$

The receiver acts first and commits to a default prediction strategy. For each default class $i \in \{1, \dots, K\}$, the receiver sets

$$\hat{y}(\emptyset) = i.$$

If features are revealed, the receiver uses the Bayes classifier conditioned on the fact that the sender is not of type i :

$$\hat{y}(x) = \arg \max_{c \in \{1, \dots, K\}} P(Y = c \mid X = x, U \neq i).$$

Define

$$\pi_i := P(U = i), \quad r_X^{(-i)} := E[\mathbf{1}\{\hat{y}(X) \neq Y\} \mid U \neq i].$$

The quantity $r_X^{(-i)}$ is the Bayes risk on the subpopulation with preference $U \neq i$.

4.2 Sender Best Response

Lemma 4.1 (Best response). *For a committed default class i , the sender withholds features if $U = i$ and reveals features if $U \neq i$.*

Proof. The sender minimizes its own 0-1 loss, namely

$$E[\mathbf{1}\{\hat{y}(M) \neq U\}].$$

We consider the two possible sender types separately.

Case 1: $U = i$. If the sender withholds, then $M = \emptyset$ and the receiver predicts $\hat{y}(\emptyset) = i$. The sender's loss is therefore

$$\mathbf{1}\{i \neq i\} = 0.$$

This is the minimum possible loss, so withholding is optimal.

Case 2: $U \neq i$. If the sender withholds, the receiver again predicts i , so the sender's loss is

$$\mathbf{1}\{i \neq U\} = 1.$$

This is the worst possible 0-1 loss. If the sender reveals, then with probability $1 - q$ the features survive and the receiver may output a class closer to U , so the sender can strictly improve relative to withholding. Hence revealing is optimal for $U \neq i$.

Therefore the sender's action partitions the population exactly into the two sets $\{U = i\}$ and $\{U \neq i\}$. $\square$

4.3 Strategic Risk

Lemma 4.2 (Strategic risk for default i). *For any default class i , the receiver's strategic risk is*

$$R_{S_i}(q) = \pi_i P(Y \neq i \mid U = i) + (1 - \pi_i) \left(q P(Y \neq i \mid U \neq i) + (1 - q) r_X^{(-i)} \right).$$

Equivalently,

$$R_{S_i}(q) = P(Y \neq i) + (1 - \pi_i)(1 - q) \left(r_X^{(-i)} - P(Y \neq i \mid U \neq i) \right).$$

Proof. By the law of total expectation, we decompose the receiver's error over the two sender types identified by the best-response lemma.

Step 1: the withholding subpopulation $U = i$. For this group, the sender always withholds, so the receiver always predicts i . The contribution to risk is

$$\pi_i P(Y \neq i \mid U = i).$$

Step 2: the revealing subpopulation $U \neq i$. For this group, the sender always reveals. The channel erases the features with probability q , in which case the receiver still predicts i and the error is $P(Y \neq i \mid U \neq i)$. With probability $1 - q$, the features are received and the receiver uses the Bayes classifier on the revealed message, incurring conditional Bayes risk $r_X^{(-i)}$.

Thus the contribution from this subpopulation is

$$(1 - \pi_i) \left(q P(Y \neq i \mid U \neq i) + (1 - q) r_X^{(-i)} \right).$$

Adding the two contributions gives the first formula.

To obtain the second formula, use the law of total probability:

$$P(Y \neq i) = \pi_i P(Y \neq i \mid U = i) + (1 - \pi_i) P(Y \neq i \mid U \neq i).$$

Substituting this identity into the first expression and grouping terms yields

$$R_{S_i}(q) = P(Y \neq i) + (1 - \pi_i)(1 - q) \left(r_X^{(-i)} - P(Y \neq i \mid U \neq i) \right).$$

$\square$

4.4 Structural Results

The previous lemma shows that for every fixed default class i , the strategic risk is an affine function of the channel noise parameter q .

4.4.1 Endpoints

At the extreme $q = 1$, every revealed message is destroyed by the channel. The receiver always falls back to the default class i , so

$$R_{S_i}(1) = P(Y \neq i).$$

Thus the optimal default at $q = 1$ is the vanilla best constant

$$c^* \in \arg \min_{i \in \{1, \dots, K\}} P(Y \neq i) = \arg \max_{i \in \{1, \dots, K\}} P(Y = i).$$

At $q = 0$, the channel never erases a revealed feature. Then

$$R_{S_i}(0) = P(Y \neq i) + (1 - \pi_i) \left(r_X^{(-i)} - P(Y \neq i \mid U \neq i) \right).$$

The optimal default can depend on the informativeness of the features for each subpopulation.

Theorem 4.3 (Dominance of the vanilla best constant). *If the vanilla best constant c^* is optimal at both $q = 1$ and $q = 0$, then it is optimal for all $q \in [0, 1]$.*

Proof. Fix any competitor default $j \neq c^*$ and define the risk difference

$$\Delta_j(q) := R_{S_j}(q) - R_{S_{c^*}}(q).$$

Because both $R_{S_j}(q)$ and $R_{S_{c^*}}(q)$ are affine functions of q , the difference $\Delta_j(q)$ is also affine.

By assumption, c^* is optimal at both endpoints, so

$$\Delta_j(1) \geq 0 \quad \text{and} \quad \Delta_j(0) \geq 0.$$

An affine function on $[0, 1]$ attains its minimum at an endpoint. Therefore $\Delta_j(q) \geq 0$ for all $q \in [0, 1]$.

Since this holds for every j , we conclude that $R_{S_{c^*}}(q) \leq R_{S_j}(q)$ for all competitors and all $q \in [0, 1]$. Hence c^* is optimal throughout the interval. $\square$

Theorem 4.4 (Piecewise-linear partition of the unit interval). *The optimal strategic risk*

$$R_S^*(q) = \min_{i \in \{1, \dots, K\}} R_{S_i}(q)$$

is a piecewise-linear, concave function of q . Consequently, the interval $[0, 1]$ can be partitioned into at most K subintervals, and on each subinterval a single default class is optimal.

Proof. Each $R_{S_i}(q)$ is affine in q , so the pointwise minimum of the finite collection $\{R_{S_i}(q)\}_{i=1}^K$ is the lower envelope of K lines.

The lower envelope of finitely many affine functions is always piecewise linear and concave. Because there are only K lines, there can be at most K linear segments in the envelope and therefore at most $K - 1$ transition points where the active default class changes.

Hence $[0, 1]$ is partitioned into at most K subintervals, and on each interval one default class uniquely determines the optimum risk. $\square$

4.5 Computing the Partition

The piecewise-linear structure gives a direct algorithm for finding the optimal default regions.

1. For every pair of defaults $i < j$, solve the affine equation

$$R_{S_i}(q) = R_{S_j}(q)$$

to obtain a candidate intersection point.

2. Keep only the intersections that lie strictly in $(0, 1)$.
3. Sort the valid breakpoints as

$$0 = q_0 < q_1 < \cdots < q_m < q_{m+1} = 1.$$

4. For each interval $[q_{k-1}, q_k]$, evaluate the risks at a midpoint and choose the default with minimum risk there.

This procedure identifies the exact default class that is optimal on each region of q .

4.6 Conclusion

The multiclass decision problem reduces to a clean lower-envelope geometry. Each default class induces an affine strategic-risk line in the channel-noise parameter, the sender's best response partitions the population by preference, and the receiver's optimal policy is obtained by comparing finitely many lines. The result is a piecewise-linear partition of $[0, 1]$ with at most K regions.

Chapter 5

Multiclassification Learning

Abstract

We study the batch-learning problem for strategic multiclass voluntary disclosure under 0-1 loss. For each default class, the receiver trains a multiclass linear predictor on the subpopulation that would reveal features under that default, evaluates the resulting strategic risk, and then selects the best default-predictor pair. This chapter proves the sample-size concentration needed for the revealed subpopulation, derives uniform convergence for a fixed default, combines the bounds over all defaults, and concludes with an excess-risk guarantee for the joint learning algorithm.

5.1 Learning Setup

Let $X \in \mathbb{R}^d$ and $U, Y \in \{1, \dots, K\}$. We observe an i.i.d. sample

$$S = \{(u_j, x_j, y_j)\}_{j=1}^n$$

drawn from the distribution of (U, X, Y) .

For each default class $i \in \{1, \dots, K\}$, the receiver commits to a multiclass linear predictor with weight matrix

$$\Theta_i \in \mathbb{R}^{K \times d}, \quad \|\Theta_i\|_\infty \leq C,$$

whose c -th row is $\theta_{i,c}^\top$. The predictor computes the score vector

$$s = \Theta_i x, \quad s_c = \langle \theta_{i,c}, x \rangle,$$

and outputs

$$h_{\Theta_i}(x) = \arg \max_{c \in \{1, \dots, K\}} \langle \theta_{i,c}, x \rangle.$$

If no feature vector is observed, the receiver predicts the default class i . If the sender's preference is not i , the sender reveals the feature vector; otherwise the sender withholds it. Thus, for default i , the effective training data for the predictor come from the random subpopulation $\{j : u_j \neq i\}$.

Define the hypothesis class and its 0-1 loss class by

$$\mathcal{H}_i = \{h_{\Theta_i} : \|\Theta_i\|_\infty \leq C\}, \quad \mathcal{L}_i = \{(x, y) \mapsto \mathbf{1}\{h(x) \neq y\} : h \in \mathcal{H}_i\}.$$

For a fixed default i and predictor h , the strategic risk is

$$R_{S_i}(h, q) = \pi_i P(Y \neq i \mid U = i) + (1 - \pi_i) \left(q P(Y \neq i \mid U \neq i) + (1 - q) R_i(h) \right),$$

where

$$\pi_i = P(U = i), \quad R_i(h) = P(h(X) \neq Y \mid U \neq i).$$

The empirical strategic risk is

$$\hat{R}_{S_i}(h, q) = \frac{1}{n} \sum_{j=1}^n \left(\mathbf{1}\{u_j = i\} \mathbf{1}\{i \neq y_j\} + \mathbf{1}\{u_j \neq i\} (q \mathbf{1}\{i \neq y_j\} + (1 - q) \mathbf{1}\{h(x_j) \neq y_j\}) \right).$$

5.2 Joint Learning Algorithm

For each default i , define the revealed-subpopulation sample size

$$n_i = |\{j : u_j \neq i\}|.$$

The joint learning algorithm proceeds as follows.

1. For each default i , train the empirical predictor on the revealed subpopulation:

$$\hat{\Theta}_i = \arg \min_{\|\Theta\|_\infty \leq C} \frac{1}{n_i} \sum_{j: u_j \neq i} \mathbf{1}\{h_\Theta(x_j) \neq y_j\}.$$

2. Evaluate the corresponding strategic risk $\hat{R}_{S_i}(h_{\hat{\Theta}_i}, q)$.

3. Select the best combined strategy

$$(\hat{i}^*, \hat{\Theta}_{\hat{i}^*}) = \arg \min_{i \in \{1, \dots, K\}} \hat{R}_{S_i}(h_{\hat{\Theta}_i}, q).$$

The learning problem is to show that the risk of $(\hat{i}^*, \hat{\Theta}_{\hat{i}^*})$ converges to the optimal strategic risk in hindsight.

5.3 Effective Sample Size

Assumption 5.1 (Nondegenerate revealed mass). *There exists a constant $\gamma > 0$ such that*

$$\min_{i \in \{1, \dots, K\}} P(U \neq i) \geq \gamma.$$

Lemma 5.2 (Effective sample size). *With probability at least $1 - \delta/2$, the revealed-subpopulation sample size satisfies*

$$n_i \geq \frac{\gamma n}{2}$$

for all $i \in \{1, \dots, K\}$ simultaneously, provided n is large enough.

Proof. Fix a default i . The indicator variables $\mathbf{1}\{u_j \neq i\}$ are i.i.d. Bernoulli random variables with mean

$$1 - \pi_i = P(U \neq i) \geq \gamma.$$

Therefore

$$n_i = \sum_{j=1}^n \mathbf{1}\{u_j \neq i\}$$

is Binomial($n, 1 - \pi_i$) and

$$\mathbb{E}[n_i] = n(1 - \pi_i) \geq n\gamma.$$

By Hoeffding's inequality,

$$P\left(\frac{n_i}{n} - \mathbb{E}\left[\frac{n_i}{n}\right] \leq -\frac{\gamma}{2}\right) \leq \exp\left(-2n\left(\frac{\gamma}{2}\right)^2\right) = \exp\left(-\frac{n\gamma^2}{2}\right).$$

Equivalently,

$$P\left(n_i < \frac{\gamma n}{2}\right) \leq \exp\left(-\frac{n\gamma^2}{2}\right).$$

Applying a union bound over the K defaults gives

$$P\left(\exists i : n_i < \frac{\gamma n}{2}\right) \leq K \exp\left(-\frac{n\gamma^2}{2}\right).$$

For sufficiently large n , this is at most $\delta/2$. Hence, with probability at least $1 - \delta/2$, every default receives $\Omega(n)$ revealed samples. $\square$

5.4 Fixed-Default Generalization

We first prove a combinatorial bound on the number of possible multiclass labelings a hypothesis class can generate. This is Natarajan's extension of the Sauer-Shelah lemma.

Lemma 5.3 (Natarajan's Lemma). *Let $\mathcal{H}$ be a hypothesis class of functions from a domain to $\{1, \dots, K\}$ with Natarajan dimension d_N . For any sample S of size n , the number of realizable labelings $|\mathcal{H}|_S$ is bounded by*

$$|\mathcal{H}|_S \leq \Phi_K(d_N, n) \triangleq \sum_{i=0}^{d_N} \binom{n}{i} \binom{K}{2}^i.$$

Proof. We proceed by induction on n and d_N . For $n = 0$ or $d_N = 0$, the bound holds trivially since $\Phi_K(d_N, 0) = 1$ and $\Phi_K(0, n) = 1$. Assume the bound holds for samples of size $n - 1$. Let $S = \{x_1, \dots, x_n\}$.

Partition $\mathcal{H}|_S$ by their labelings on $S' = S \setminus \{x_n\}$. For each labeling $y \in \mathcal{H}|_{S'}$, let $C(y)$ be the set of possible labels for x_n such that the combined labeling is in $\mathcal{H}|_S$. The size of $\mathcal{H}|_S$ is exactly $\sum_{y \in \mathcal{H}|_{S'}} |C(y)|$. We can rewrite this sum as:

$$|\mathcal{H}|_S = \sum_{y \in \mathcal{H}|_{S'}} 1 + \sum_{y \in \mathcal{H}|_{S'}} (|C(y)| - 1).$$

The first term is exactly $|\mathcal{H}|_{S'}$, which by the inductive hypothesis is at most $\Phi_K(d_N, n - 1)$. For the second term, we observe that for any set of size $k \geq 1$, $k - 1 \leq \sum_{a < b} \mathbf{1}\{a \in C(y) \text{ and } b \in C(y)\}$. Thus,

$$\sum_{y \in \mathcal{H}|_{S'}} (|C(y)| - 1) \leq \sum_{a < b} \sum_{y \in \mathcal{H}|_{S'}} \mathbf{1}\{a, b \in C(y)\}.$$

For any fixed pair of labels (a, b) , the inner sum counts the number of prefix labelings y that can be extended by either a or b at x_n . Let $\mathcal{H}_{a,b}$ denote this set of prefix labelings. If a subset $T \subseteq S'$ is N-shattered by $\mathcal{H}_{a,b}$, then $T \cup \{x_n\}$ is N-shattered by $\mathcal{H}|_S$ (using the same shattering patterns on T , and using a and b for x_n). Therefore, the Natarajan dimension of $\mathcal{H}_{a,b}$ is at most $d_N - 1$.

Applying the inductive hypothesis to $\mathcal{H}_{a,b}$ for each of the $\binom{K}{2}$ pairs (a, b) , we get that the second term is bounded by $\binom{K}{2} \Phi_K(d_N - 1, n - 1)$. Putting it all together:

$$|\mathcal{H}|_S \leq \Phi_K(d_N, n - 1) + \binom{K}{2} \Phi_K(d_N - 1, n - 1).$$

Using Pascal's identity $\binom{n-1}{i} + \binom{n-1}{i-1} = \binom{n}{i}$, we obtain:

$$\begin{aligned}
|\mathcal{H}|_S &\leq \sum_{i=0}^{d_N} \binom{n-1}{i} \binom{K}{2}^i + \binom{K}{2} \sum_{j=0}^{d_N-1} \binom{n-1}{j} \binom{K}{2}^j \\
&= \sum_{i=0}^{d_N} \binom{n-1}{i} \binom{K}{2}^i + \sum_{i=1}^{d_N} \binom{n-1}{i-1} \binom{K}{2}^i \\
&= 1 + \sum_{i=1}^{d_N} \left[\binom{n-1}{i} + \binom{n-1}{i-1} \right] \binom{K}{2}^i \\
&= \sum_{i=0}^{d_N} \binom{n}{i} \binom{K}{2}^i = \Phi_K(d_N, n).
\end{aligned}$$

This completes the induction. $\square$

Lemma 5.4 (Fixed-default uniform convergence). *For a fixed default i , with probability at least $1 - \delta/K$,*

$$\sup_{h \in \mathcal{H}_i} |R_{S_i}(h, q) - \hat{R}_{S_i}(h, q)| = O\left(\sqrt{\frac{Kd \log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right).$$

In particular, the same bound holds for the empirical minimizer $\hat{h}_i = h_{\hat{\Theta}_i}$.

Proof. The strategic risk decomposes into a default-dependent constant plus the revealed-subpopulation classification risk. Since the default terms do not depend on h , it is enough to control the revealed-subpopulation loss class

$$\mathcal{L}_i = \{(x, y) \mapsto \mathbf{1}\{h(x) \neq y\} : h \in \mathcal{H}_i\}.$$

By symmetrization and the standard Rademacher generalization bound for bounded losses, with probability at least $1 - \delta/K$,

$$\sup_{h \in \mathcal{H}_i} |R_{S_i}(h, q) - \hat{R}_{S_i}(h, q)| \leq 2 \text{Rad}_{n_i}(\mathcal{L}_i) + O\left(\sqrt{\frac{\log(K/\delta)}{n_i}}\right).$$

To bound the empirical Rademacher complexity $\text{Rad}_{n_i}(\mathcal{L}_i)$, we consider the projection of the loss class $\mathcal{L}_i$ onto the n_i empirical samples. Since the loss takes values in $\{0, 1\}$, the empirical loss vectors lie in the hypercube $\{0, 1\}^{n_i}$, yielding a maximum ℓ_2 norm of $\sqrt{n_i}$. By Massart's lemma, the empirical Rademacher complexity is bounded by

$$\text{Rad}_{n_i}(\mathcal{L}_i) \leq \frac{\sqrt{n_i} \sqrt{2 \log |\mathcal{L}_i|_{S_{n_i}}}}{n_i} = \sqrt{\frac{2 \log |\mathcal{L}_i|_{S_{n_i}}}{n_i}},$$

where $|\mathcal{L}_i|_{S_{n_i}}$ is the growth function (the number of distinct loss vectors realizable on the sample). Since the loss is deterministic given the hypothesis, the number of distinct loss vectors is at most the number of distinct labelings the hypothesis class $\mathcal{H}_i$ can produce on the n_i points. A multiclass linear predictor with K rows and d coordinates per row has Natarajan dimension $d_N = O(Kd)$.

To bound the growth function, we apply Natarajan's extension of the Sauer-Shelah lemma. For a hypothesis class taking values in K categories with Natarajan dimension d_N , the number of possible labelings on n_i points is bounded by

$$|\mathcal{H}_i|_{S_{n_i}} \leq \sum_{j=0}^{d_N} \binom{n_i}{j} \binom{K}{2}^j \leq \left(\frac{en_i K^2}{2d_N}\right)^{d_N}.$$

Taking the logarithm yields $\log |\mathcal{L}_i|_{S_{n_i}} \leq \log |\mathcal{H}_i|_{S_{n_i}} = O(d_N \log(n_i K)) = O(K d \log n_i)$. Substituting this into Massart's bound gives

$$\text{Rad}_{n_i}(\mathcal{L}_i) = O\left(\sqrt{\frac{K d \log n_i}{n_i}}\right).$$

By the effective sample-size lemma, $n_i \geq \gamma n/2$ with high probability. Thus $\log n_i \leq \log n$ and $1/n_i = O(1/n)$. Substituting this into the symmetrization bound yields

$$\sup_{h \in \mathcal{H}_i} |R_{S_i}(h, q) - \hat{R}_{S_i}(h, q)| = O\left(\sqrt{\frac{K d \log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right).$$

Since $\hat{h}_i \in \mathcal{H}_i$, the same inequality holds for the empirical minimizer. $\square$

5.5 Simultaneous Bound

Lemma 5.5 (Uniform convergence over all defaults). *With probability at least $1 - \delta$,*

$$\sup_{i \in \{1, \dots, K\}} \sup_{h \in \mathcal{H}_i} |R_{S_i}(h, q) - \hat{R}_{S_i}(h, q)| = O\left(\sqrt{\frac{K d \log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right).$$

Proof. Let E_i be the event that the fixed-default bound from the previous lemma holds for default i . By that lemma, we can choose the failure probability for each default as δ/K . Therefore

$$P(E_i) \geq 1 - \frac{\delta}{K}.$$

Applying the union bound,

$$P\left(\bigcap_{i=1}^K E_i\right) \geq 1 - \sum_{i=1}^K P(E_i^c) \geq 1 - K \cdot \frac{\delta}{K} = 1 - \delta.$$

Thus, with probability at least $1 - \delta$, the fixed-default bound holds simultaneously for every default class i . $\square$

5.6 Excess Risk of the Joint Learner

Let

$$R_S^*(q) = \min_{i \in \{1, \dots, K\}} \min_{h \in \mathcal{H}_i} R_{S_i}(h, q)$$

denote the optimal strategic risk in hindsight. Let

$$(i^*, h_{i^*}^*) \in \arg \min_{i, h \in \mathcal{H}_i} R_{S_i}(h, q)$$

be an oracle optimal pair.

Theorem 5.6 (Excess risk of the learned strategy). *With probability at least $1 - \delta$,*

$$R_{S_{\hat{i}^*}}(\hat{h}_{\hat{i}^*}, q) - R_S^*(q) = O\left(\sqrt{\frac{K d \log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right).$$

In particular, this implies the looser source-level rate

$$O\left(K \sqrt{\frac{d \log n}{n}}\right).$$

Proof. On the high-probability event from the simultaneous bound, every default class satisfies

$$|R_{S_i}(h, q) - \hat{R}_{S_i}(h, q)| \leq \epsilon_n \quad \text{for all } h \in \mathcal{H}_i,$$

where

$$\epsilon_n = O\left(\sqrt{\frac{Kd \log n}{n}} + \sqrt{\frac{\log(K/\delta)}{n}}\right).$$

The empirical learner selects

$$(\hat{i}^*, \hat{h}_{\hat{i}^*}) \in \arg \min_{i \in \{1, \dots, K\}} \hat{R}_{S_i}(\hat{h}_i, q),$$

so for every competitor (i, h) ,

$$\hat{R}_{S_{\hat{i}^*}}(\hat{h}_{\hat{i}^*}, q) \leq \hat{R}_{S_i}(h, q).$$

In particular, for the oracle pair $(i^*, h_{i^*}^*)$,

$$\hat{R}_{S_{\hat{i}^*}}(\hat{h}_{\hat{i}^*}, q) \leq \hat{R}_{S_{i^*}}(h_{i^*}^*, q).$$

Add and subtract empirical risks:

$$\begin{aligned} R_{S_{\hat{i}^*}}(\hat{h}_{\hat{i}^*}, q) - R_S^*(q) &= (R_{S_{\hat{i}^*}}(\hat{h}_{\hat{i}^*}, q) - \hat{R}_{S_{\hat{i}^*}}(\hat{h}_{\hat{i}^*}, q)) \\ &+ (\hat{R}_{S_{\hat{i}^*}}(\hat{h}_{\hat{i}^*}, q) - \hat{R}_{S_{i^*}}(h_{i^*}^*, q)) + (\hat{R}_{S_{i^*}}(h_{i^*}^*, q) - R_{S_{i^*}}(h_{i^*}^*, q)). \end{aligned}$$

The middle term is nonpositive by empirical optimality, so

$$R_{S_{\hat{i}^*}}(\hat{h}_{\hat{i}^*}, q) - R_S^*(q) \leq 2\epsilon_n.$$

Substituting the expression for ϵ_n gives the stated rate.

Finally, because $\sqrt{K} \leq K$ for all $K \geq 1$, the sharper bound above implies the looser source-style rate

$$O\left(K\sqrt{\frac{d \log n}{n}}\right).$$

□

5.7 Conclusion

The multiclass learning problem follows the same strategic pattern as the regression learning problem, but with a finite family of default classes. The key proof steps are: control the random revealed sample size, establish uniform convergence for each default via VC and Rademacher bounds, union bound over the K defaults, and then use the empirical optimality of the selected pair to obtain the final excess-risk guarantee.

Source Note

The regression chapters are adapted from the provided PDFs in `Sources/`. In particular, they draw on `strategic_revelation.pdf` and `main.pdf`. The multiclass chapters are original extensions written to match the same style and scope.